

# Influence-Aware Financial Sentiment Analysis for Stock Trend Prediction Using FinBERT and Social Network Analysis

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**Abstract**—Stock market prediction remains a difficult task because market movements are shaped by numerical indicators, investor behavior, public expectations, and information diffusion. Social media platforms such as Twitter provide large-scale real-time information about investor sentiment, but most sentiment-based forecasting approaches aggregate all users equally. This assumption is weak because online financial communities are highly unequal: a small number of influential users can dominate information flow while most users remain peripheral.

This paper presents an influence-aware stock trend prediction framework that integrates Financial Natural Language Processing, Social Network Analysis, and Machine Learning. A two-stage FinBERT adaptation strategy is used for sentiment classification. In the first stage, VADER-generated weak labels are used to adapt FinBERT to financial Twitter language. In the second stage, the weakly adapted model is further fine-tuned using manually annotated gold-standard labels. The final model achieves 71.54% accuracy and 71.40% weighted F1-score and is applied to approximately 3.6 million company-related tweets.

To model user influence, a Twitter interaction network is constructed and PageRank centrality scores are calculated. These scores are combined with tweet-level sentiment predictions to produce PageRank-weighted sentiment features. The sentiment features are then merged with historical stock market indicators and evaluated using multiple machine learning models across six NASDAQ stocks: AAPL, AMZN, GOOG, GOOGL, MSFT, and TSLA. The results show that sentiment improves prediction performance for AAPL and TSLA, provides moderate gains for AMZN and MSFT, and has mixed effects for GOOG and GOOGL. Statistical testing indicates significant improvement mainly for AAPL. These findings suggest that influence-aware sentiment can enhance financial forecasting, but its usefulness is company-dependent rather than universal.

**Index Terms**—Financial NLP, FinBERT, Sentiment Analysis, Social Network Analysis, PageRank, Stock Trend Prediction, Machine Learning, Explainable AI.

## I. INTRODUCTION

Financial markets are influenced not only by historical prices and technical indicators, but also by investor expectations, public reactions, and the speed at which information spreads. Twitter has become an important platform where investors, analysts, journalists, companies, and retail traders discuss market events in real time. These discussions create a large and dynamic source of public sentiment that may help explain or anticipate stock market movements.

Traditional stock forecasting models mainly use price-based features such as open price, close price, volume, moving

averages, momentum, and volatility indicators. Although such variables capture numerical market behavior, they do not directly represent market psychology, investor mood, or public reactions to company events. Recent advances in Natural Language Processing have made it possible to extract sentiment from unstructured financial text, especially using transformer-based models such as BERT and FinBERT.

However, sentiment extraction alone is not sufficient. Most sentiment-based forecasting systems assume that every tweet contributes equally to market sentiment. This is unrealistic. Social networks are characterized by unequal visibility and influence. Some users receive more engagement, are retweeted by other important accounts, and can amplify financial narratives more effectively than ordinary users. Therefore, the source of a sentiment signal may be as important as the sentiment polarity itself.

This study investigates whether stock trend prediction can be improved by combining financial sentiment analysis with PageRank-based user influence modeling. The prediction task is formulated as a binary classification problem: predicting whether the next trading day's opening price will move up or down. The study addresses three research questions:

- RQ1: Can a two-stage FinBERT adaptation strategy improve financial sentiment classification?
- RQ2: Does PageRank-weighted sentiment improve stock trend prediction compared with raw sentiment and price-only features?
- RQ3: Are the observed improvements statistically significant across different companies and models?

The main contributions are:

- A semi-supervised two-stage FinBERT fine-tuning pipeline using weak VADER labels and manually annotated gold labels.
- Large-scale sentiment inference on approximately 3.6 million company-related financial tweets.
- Construction of a Twitter interaction network with 1.6M+ nodes and 1.7M+ edges.
- Design of PageRank-weighted sentiment features that combine textual polarity with user influence.
- Comparative evaluation of nine machine learning models under three feature settings across six NASDAQ stocks.

- Statistical testing of the effectiveness of sentiment-enhanced prediction models.

## II. RELATED WORK

### A. Financial Sentiment Analysis

Financial sentiment analysis has evolved from simple lexicon-based methods to machine learning, deep learning, and transformer-based approaches. Early work used dictionaries and rule-based methods to estimate polarity from financial text. VADER is particularly popular for social media because it handles punctuation, capitalization, intensifiers, and short informal expressions. However, VADER is still lexicon-based and may struggle with financial terminology and domain-specific meanings.

Machine learning approaches such as Support Vector Machines, Random Forests, and Naive Bayes improved sentiment classification by learning patterns from labeled data. Later, deep learning methods such as recurrent neural networks and LSTM architectures improved the modeling of contextual and temporal patterns. Transformer models further improved text classification by learning contextual word representations.

FinBERT is especially relevant because it is adapted to financial language. Unlike general BERT, FinBERT is trained or fine-tuned on financial corpora such as analyst reports, earnings calls, and financial news. This makes it more suitable for identifying positive, neutral, and negative sentiment in financial text. Since tweets about stocks contain ticker symbols, finance-specific terms, abbreviations, and informal language, FinBERT provides a strong foundation for sentiment extraction.

### B. Social Network Analysis and Influence

Social Network Analysis studies relationships between actors and the flow of information through networks. In Twitter data, users can be represented as nodes, while retweets, mentions, replies, or engagement relations can be represented as directed edges. Centrality measures are used to identify influential users within these networks.

Degree centrality counts direct connections but does not distinguish between connections from important and unimportant users. Betweenness centrality identifies users who connect different communities but does not necessarily measure the authority of a user's content. PageRank is more suitable for this study because it considers both the number and quality of incoming links. A user receives a high PageRank score when they are connected to other influential users. This makes PageRank useful for modeling authority and influence in financial conversations.

Prior studies suggest that network structure can provide additional information beyond textual sentiment. In financial social media, influential users may amplify market narratives and shape investor attention. Therefore, weighting sentiment by influence can reduce noise from low-impact users and emphasize signals from users who occupy central positions in the information network.

### C. Stock Trend Prediction

Stock trend prediction has traditionally relied on statistical and econometric methods. More recent approaches use machine learning and deep learning models to capture non-linear patterns in price data. Common algorithms include Logistic Regression, SVM, Random Forest, Gradient Boosting, XGBoost, LightGBM, CatBoost, and neural networks.

Many studies show that combining technical indicators with external information such as news or social media sentiment can improve prediction. However, the effectiveness of sentiment is not uniform. It may depend on the company, sector, market period, social media activity, and the quality of the sentiment signal. This study contributes to the literature by testing not only raw sentiment, but also influence-weighted sentiment across multiple models and companies.

## III. DATA AND METHODOLOGY

### A. System Architecture

The framework consists of five main stages: data collection, sentiment classification, network construction, feature engineering, and stock trend prediction. Figure 1 summarizes the complete workflow.

### B. Datasets

Three datasets are used in this research. The first is the IEEE Financial Tweet Collection, containing 943,672 tweets from April 9 to July 16, 2020. This dataset includes 1,300 manually labeled tweets annotated as positive, neutral, or negative. The second dataset is a Kaggle financial tweet dataset covering 2015–2020 and containing more than 3 million tweets about major technology companies. The third dataset contains historical NASDAQ stock prices from 2010 to 2020.

TABLE I  
DATASETS USED IN THE STUDY

| Dataset                  | Purpose                                              | Time Period  |
|--------------------------|------------------------------------------------------|--------------|
| IEEE financial tweets    | Gold labels and weak-label training data             | Apr–Jul 2020 |
| Kaggle company tweets    | Large-scale sentiment inference and network analysis | 2015–2020    |
| NASDAQ historical prices | Stock trend prediction and validation                | 2010–2020    |

The stock prediction experiments focus on AAPL, AMZN, GOOG, GOOGL, MSFT, and TSLA. Historical market variables include open price, close price, high price, low price, and trading volume.

### C. Tweet Preprocessing

Tweets contain noise such as URLs, mentions, hashtags, emojis, punctuation, numbers, cashtags, and informal expressions. The preprocessing pipeline includes lowercasing, URL removal, mention handling, hashtag normalization, punctuation removal, symbol replacement, financial abbreviation expansion, whitespace normalization, and dataset-specific stop-word removal.

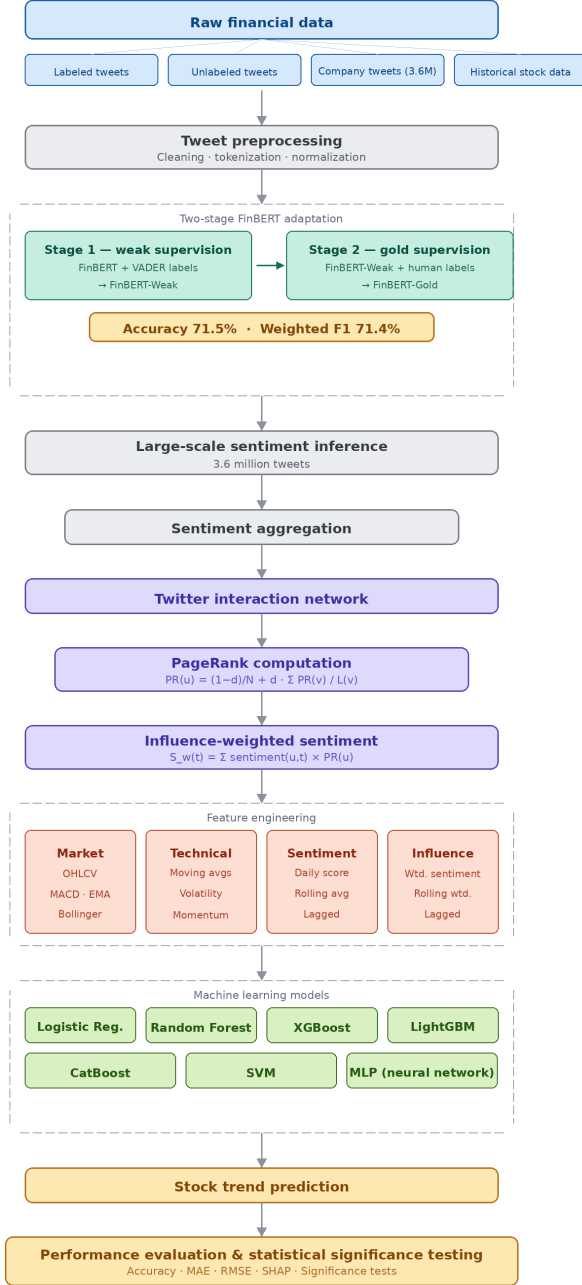


Fig. 1. Overall research framework integrating Financial NLP, PageRank-based influence modeling, feature engineering, and stock trend prediction.

|   | text                                               | Text_Cleaned                                      |
|---|----------------------------------------------------|---------------------------------------------------|
| 0 | RT @RobertBeadles: Yo 🏆🏆🏆Enter to WIN 1,000 Mon... | robertbeadles yo enter win monarch tokens cras... |
| 1 | #SriLanka surcharge on fuel removed!🇱🇰🇱🇰The ...    | srilanka surcharge fuel removed surcharge rs i... |
| 2 | Net issuance increases to fund fiscal programs...  | net issuance increases fund fiscal programs gt... |
| 3 | RT @bentboolean: How much of Amazon's traffic ...  | bentboolean much amazons traffic served fastly... |
| 4 | \$AMD Ryzen 4000 desktop CPUs looking 'great' a... | ryzen desktop cpus looking great track launch ... |

Fig. 2. Example of raw tweet text and cleaned text after preprocessing.

Stopword removal was performed in two stages. General English stopwords were removed first. Then, financial and dataset-specific stopwords such as ticker symbols, repeated company names, platform-related terms, and promotional words were removed. This improved the quality of textual data for sentiment classification.

#### IV. TWO-STAGE FINBERT ADAPTATION

##### A. Weak and Gold Labeling Strategy

A two-stage semi-supervised strategy is used to adapt FinBERT to financial tweets. The manually labeled gold dataset contains 1,300 tweets. To expand the training data, VADER is used to weakly label 3,700 additional tweets. This produces a combined dataset of 5,000 tweets with 1,300 high-quality human labels and 3,700 weak labels.

The gold dataset contains 40.6% positive, 32.6% neutral, and 26.8% negative tweets. The weak VADER-labeled dataset has a similar distribution but contains more positive tweets.

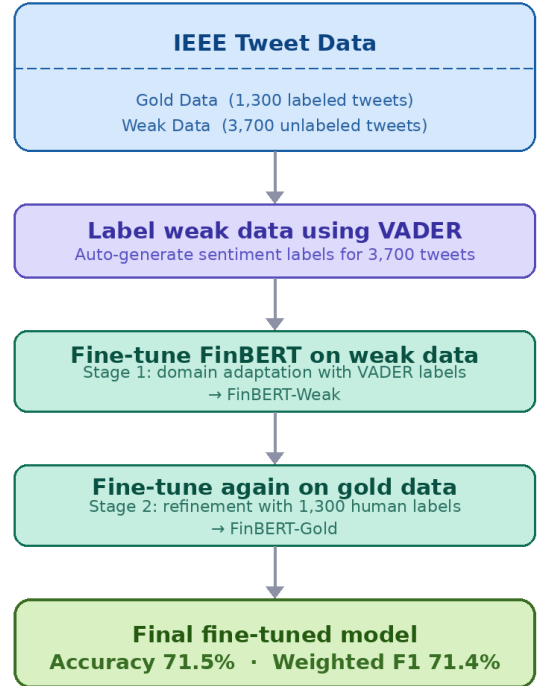


Fig. 3. Two-stage FinBERT adaptation using weak VADER labels followed by gold human-labeled data.

##### B. Stage 1: Weak Supervision

In the first stage, the weak dataset is balanced by oversampling neutral and negative classes. The FinBERT tokenizer is applied to encode the tweets, and the HuggingFace Trainer API is used for training. The model is fine-tuned for 3 epochs on the weak labels, producing FinBERT-Weak. This stage adapts the model to the language of financial tweets.

### C. Stage 2: Gold Supervision

In the second stage, FinBERT-Weak is further fine-tuned on the manually labeled gold dataset for 5 epochs. This stage corrects noise introduced by weak labeling and improves sentiment classification performance. The final model is referred to as FinBERT-Gold.

TABLE II  
FINBERT-GOLD SENTIMENT CLASSIFICATION PERFORMANCE

| Metric            | Score  |
|-------------------|--------|
| Accuracy          | 71.54% |
| Weighted F1-score | 71.40% |
| Evaluation loss   | 1.68   |

### D. Comparison With Prior Models

The final model was compared with previous models evaluated on the same S&P 500 tweet period. The proposed FinBERT-Gold model slightly outperformed the T5-based JAISIS model, achieving 71.54% accuracy compared with 70.26%.

TABLE III  
COMPARISON WITH PREVIOUS S&P 500 TWEET-BASED MODELS

| Model                 | Accuracy (%) |
|-----------------------|--------------|
| BW-SVM                | 56.38        |
| E-NN                  | 58.83        |
| WB-NN                 | 60.25        |
| WB-CNN                | 60.57        |
| E-CNN                 | 61.45        |
| EB-NN                 | 62.84        |
| EB-CNN                | 64.21        |
| SI-RCNN               | 62.03        |
| WI-RCNN               | 61.31        |
| Kamalov et al.        | 55.00        |
| T5-Based JAISIS       | 70.26        |
| Proposed FinBERT-Gold | 71.54        |

After training, FinBERT-Gold is applied to the main company tweet dataset. Since the dataset is large, it is divided into 20 chunks. Each chunk is processed separately and the outputs are merged into a single sentiment-labeled dataset.

## V. SOCIAL NETWORK ANALYSIS AND INFLUENCE MODELING

### A. Twitter Interaction Network

A Twitter interaction network is constructed to model how financial information spreads among users. The network contains both users and tweets as nodes, and directed edges represent engagements such as likes, retweets, comments, and mentions. The network is large and sparse, which is typical of social media systems.

These results indicate that most nodes have few connections, while a small core of users and tweets forms a more connected subnetwork. This supports the use of network centrality measures for identifying influential accounts.

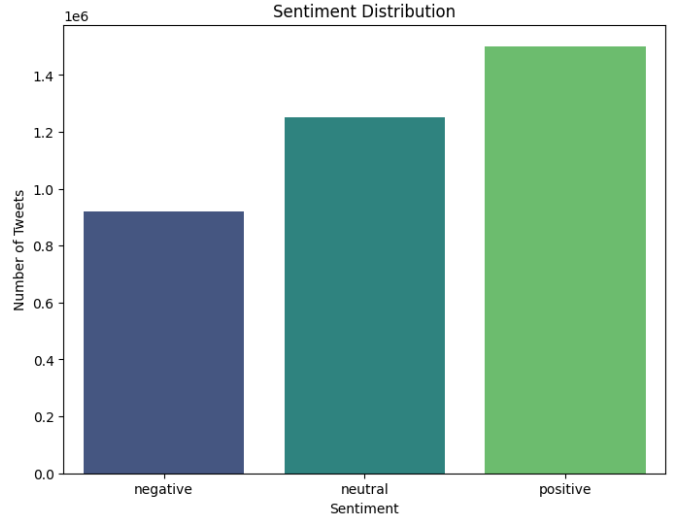


Fig. 4. Distribution of predicted sentiment categories in the large-scale company tweet dataset.

TABLE IV  
TWITTER INTERACTION NETWORK STATISTICS

| Measure                     | Value       |
|-----------------------------|-------------|
| Total nodes                 | 1,627,841   |
| Total edges                 | 1,750,309   |
| Average degree              | 2.15        |
| Density                     | 0.000001    |
| Largest connected component | 5,263 nodes |

### B. PageRank Centrality

PageRank is used to estimate user influence. It assigns higher scores to users who are referenced or engaged with by other influential users. The PageRank equation is:

$$PR(u) = \frac{1-d}{N} + d \sum_{v \in B(u)} \frac{PR(v)}{L(v)} \quad (1)$$

where  $PR(u)$  is the PageRank score of user  $u$ ,  $d$  is the damping factor,  $N$  is the total number of users,  $B(u)$  is the set of users linking to  $u$ , and  $L(v)$  is the out-degree of user  $v$ .

TABLE V  
TOP 10 INFLUENTIAL USERS BY PAGERANK

| User          | PageRank Score |
|---------------|----------------|
| RobinhoodApp  | 0.003590       |
| elonmusk      | 0.000362       |
| themotleyfool | 0.000240       |
| WSJ           | 0.000179       |
| benzinga      | 0.000122       |
| Tesla         | 0.000120       |
| jimcramer     | 0.000115       |
| IBDinvestors  | 0.000094       |
| CNBC          | 0.000054       |
| amazon        | 0.000052       |

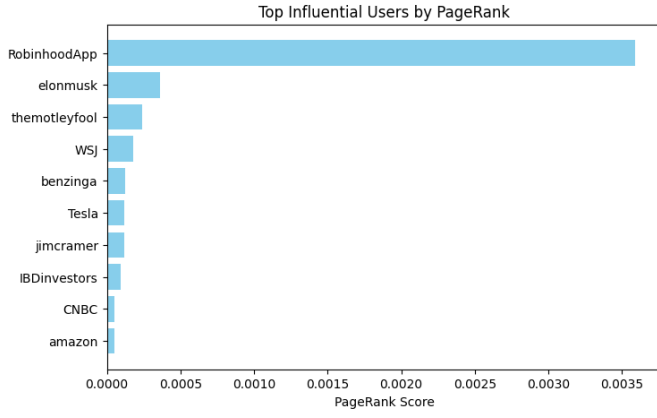


Fig. 5. Top users ranked by PageRank centrality in the Twitter interaction network.

### C. PageRank-Weighted Sentiment

Each tweet is assigned a sentiment score: positive = +1, neutral = 0, and negative = -1. This score is multiplied by the PageRank score of the tweet author:

$$\text{WeightedSentiment} = \text{Sentiment} \times \text{PageRank} \quad (2)$$

Weighted sentiment values are aggregated daily for each company. This creates sentiment indicators that reflect both market mood and the influence of the source.

## VI. FEATURE ENGINEERING AND EXPERIMENTAL DESIGN

### A. Market, Sentiment, and Influence Features

The final prediction dataset combines historical stock prices, technical indicators, raw sentiment features, and PageRank-weighted sentiment features. Technical indicators include moving averages, exponential moving average, Bollinger Bands, MACD, log momentum, price spreads, lagged prices, rolling averages, volume lags, and day-of-week features.

Sentiment features are constructed according to market timing. Tweets posted during market hours are mapped to the open-to-close period, while tweets posted after market close, before market open, or on non-trading days are mapped to the next trading day close-to-open period. The resulting features include:

- sentiment\_mapped\_open\_to\_close
- sentiment\_mapped\_close\_to\_open
- weighted\_sentiment\_open\_to\_close
- weighted\_sentiment\_close\_to\_open

Lagged and rolling-window versions of these variables are also created.

The feature-target heatmap shows that open-close spread has the strongest linear correlation with the target, while log momentum has a moderate positive relationship. Sentiment features show weaker linear correlations, but they may still be useful for non-linear models.

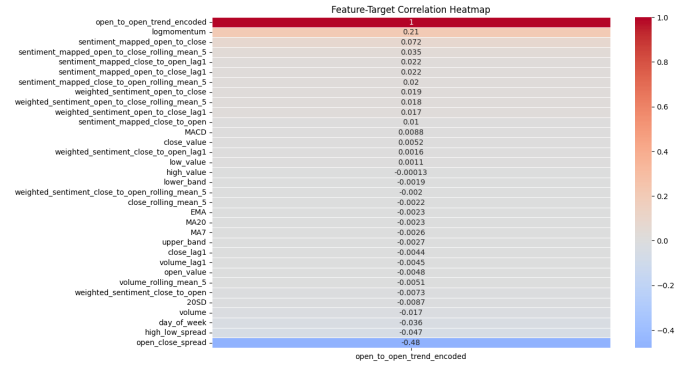


Fig. 6. Feature-target correlation heatmap showing relationships between engineered features and the open-to-open trend target.

### B. Feature Sets

Three feature sets are evaluated:

- **Price-only features:** historical prices and technical indicators.
- **Basic sentiment features:** price features plus raw sentiment variables.
- **Weighted sentiment features:** price features plus PageRank-weighted sentiment variables.

### C. Prediction Target and Evaluation

The prediction task is binary classification. The target variable, *open\_to\_open\_trend\_encoded*, indicates whether the next trading day's opening price is higher or lower than the previous opening price.

The dataset is split by time. Data from 2015 to 2018 is used for training, and data from 2019 is used for testing. This design simulates a realistic forecasting setup because future data are not used during training. Models are evaluated using accuracy, macro F1-score, and weighted F1-score.

### D. Models

Nine machine learning models are evaluated: Logistic Regression, Support Vector Machine, Random Forest, Gradient Boosting, Bagging Decision Tree, XGBoost, LightGBM, CatBoost, and Multi-Layer Perceptron.

## VII. EXPERIMENTAL RESULTS

### A. Sentiment Classification Results

The two-stage FinBERT adaptation achieved 71.54% accuracy and 71.40% weighted F1-score. This result suggests that weak-to-gold fine-tuning can effectively adapt a domain-specific financial language model to noisy social media text. The result also slightly improves over the T5-based model reported in previous work on the same tweet period.

### B. Stock Prediction Results: AAPL

For AAPL, sentiment features improved most models. The strongest gain from basic sentiment appears in MLP (+2.39%) and Gradient Boosting (+1.99%). Weighted sentiment improves Bagging, Random Forest, and XGBoost.

TABLE VI  
AAPL ACCURACY COMPARISON

| Model              | No Sent. | Basic  | Weighted |
|--------------------|----------|--------|----------|
| Bagging            | 0.7410   | 0.7530 | 0.7530   |
| CatBoost           | 0.7610   | 0.7649 | 0.7649   |
| GradientBoosting   | 0.7331   | 0.7530 | 0.7450   |
| LightGBM           | 0.7610   | 0.7689 | 0.7649   |
| LogisticRegression | 0.7649   | 0.7689 | 0.7689   |
| MLP                | 0.7490   | 0.7729 | 0.7530   |
| RandomForest       | 0.7570   | 0.7649 | 0.7689   |
| SVM                | 0.7689   | 0.7888 | 0.7689   |
| XGBoost            | 0.7689   | 0.7610 | 0.7729   |

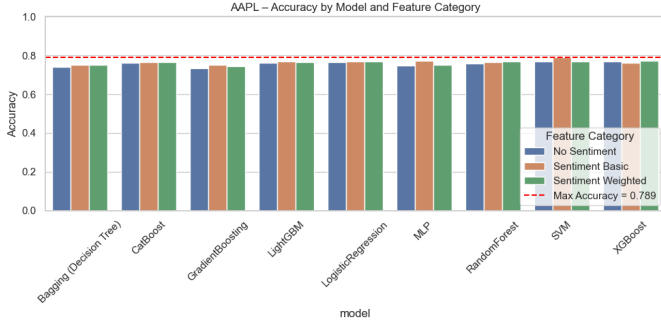


Fig. 7. AAPL accuracy comparison across feature categories.

### C. Stock Prediction Results: AMZN and MSFT

For AMZN, basic sentiment improves most models, especially SVM (+3.19%), XGBoost (+1.59%), and LightGBM (+1.20%). Weighted sentiment provides gains for selected models, particularly Random Forest (+1.59%). However, some models decline, showing that sentiment integration is model-dependent.

For MSFT, weighted sentiment is often more useful than basic sentiment. Bagging, Gradient Boosting, LightGBM, MLP, and XGBoost improve with weighted sentiment. XGBoost gains +2.79% with basic sentiment and +2.39% with weighted sentiment.

### D. Stock Prediction Results: GOOG and GOOGL

GOOG produces mixed and often negative results. Most models decline when sentiment features are added. However,

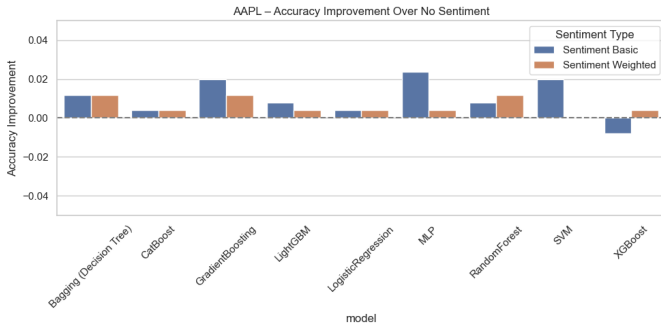


Fig. 8. AAPL accuracy improvement over the price-only baseline.

TABLE VII  
BEST SENTIMENT-ENHANCED IMPROVEMENTS FOR AMZN AND MSFT

| Ticker | Model        | Best Gain | Feature Type |
|--------|--------------|-----------|--------------|
| AMZN   | SVM          | +0.0319   | Basic        |
| AMZN   | RandomForest | +0.0159   | Weighted     |
| AMZN   | XGBoost      | +0.0159   | Basic        |
| MSFT   | XGBoost      | +0.0279   | Basic        |
| MSFT   | XGBoost      | +0.0239   | Weighted     |
| MSFT   | Bagging      | +0.0199   | Weighted     |
| MSFT   | LightGBM     | +0.0199   | Weighted     |

Gradient Boosting improves by +2.79% with weighted sentiment, and XGBoost improves by +1.59% with weighted sentiment. This suggests that sentiment can help only specific model types for GOOG.

GOOGL also shows mixed results. Logistic Regression reaches the highest observed accuracy of 0.8566 using weighted sentiment. SVM improves by +1.59% with both basic and weighted sentiment, while XGBoost improves by +1.99% with basic sentiment. However, LightGBM, MLP, and Random Forest decline with sentiment features.

### E. Stock Prediction Results: TSLA

TSLA shows the strongest benefit from weighted sentiment. XGBoost improves by +4.38%, Bagging improves by +3.19%, and Random Forest improves by +1.99% when weighted sentiment is used. This suggests that Twitter influence is particularly relevant for TSLA, likely because Tesla-related market discussions are strongly shaped by social media activity and influential accounts.

TABLE VIII  
TSLA ACCURACY COMPARISON

| Model              | No Sent. | Basic  | Weighted |
|--------------------|----------|--------|----------|
| Bagging            | 0.8287   | 0.8287 | 0.8606   |
| CatBoost           | 0.8406   | 0.8367 | 0.8486   |
| GradientBoosting   | 0.8287   | 0.8327 | 0.8446   |
| LightGBM           | 0.8247   | 0.8287 | 0.8327   |
| LogisticRegression | 0.8367   | 0.8207 | 0.8167   |
| MLP                | 0.7968   | 0.7888 | 0.7809   |
| RandomForest       | 0.8287   | 0.8287 | 0.8486   |
| SVM                | 0.8247   | 0.7928 | 0.8167   |
| XGBoost            | 0.8048   | 0.8127 | 0.8486   |

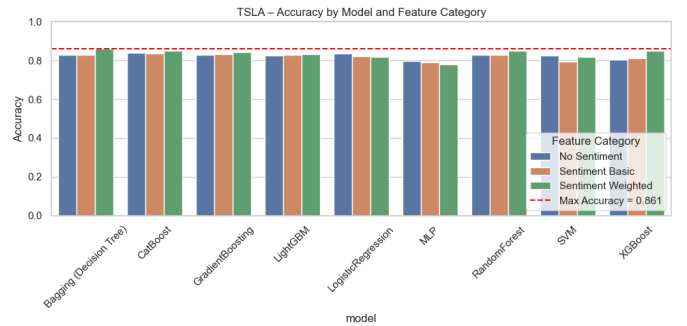


Fig. 9. TSLA accuracy comparison across feature categories.



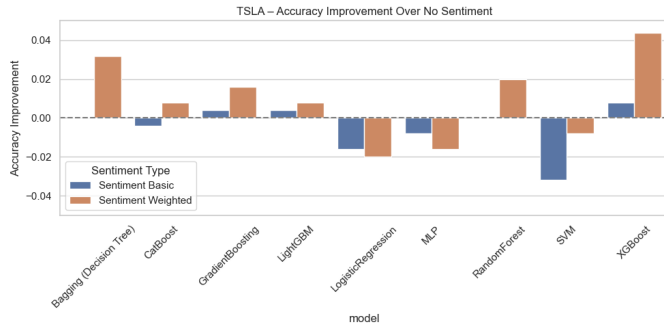


Fig. 10. TSLA accuracy improvement over the price-only baseline.

## F. Company-Level Summary

TABLE IX  
SUMMARY OF SENTIMENT CONTRIBUTION BY COMPANY

| Company | Effect   | Interpretation            |
|---------|----------|---------------------------|
| AAPL    | Strong   | Significant improvement   |
| TSLA    | Strong   | Weighted sentiment useful |
| AMZN    | Moderate | Model-dependent gains     |
| MSFT    | Moderate | Selected improvements     |
| GOOG    | Mixed    | Weak consistency          |
| GOOGL   | Mixed    | Feature-dependent effect  |

## VIII. STATISTICAL SIGNIFICANCE TESTING

To determine whether performance changes are statistically meaningful, paired t-tests and Wilcoxon signed-rank tests are applied to compare F1-scores from sentiment-enhanced models against models without sentiment features. Cohen’s  $d$  is used to measure effect size.

TABLE X  
PAIRED TESTS FOR SENTIMENT FEATURE EFFECTIVENESS

| Ticker | Mean Diff. | t-test p | Wilcoxon p | Cohen’s d |
|--------|------------|----------|------------|-----------|
| AAPL   | 0.006276   | 0.0026   | 0.0039     | 1.433     |
| MSFT   | 0.007688   | 0.1424   | 0.1289     | 0.542     |
| TSLA   | 0.009184   | 0.2308   | 0.2500     | 0.432     |
| GOOGL  | 0.000953   | 0.8711   | 0.8203     | 0.056     |
| AMZN   | -0.000936  | 0.7951   | 0.8203     | -0.089    |
| GOOG   | -0.012383  | 0.1374   | 0.1641     | -0.550    |

Only AAPL shows statistically significant improvement, with p-values below 0.01 and a large effect size. For the remaining tickers, improvements are not statistically significant. This is an important result because it prevents overclaiming. The findings suggest that sentiment can improve prediction, but its effect is not universal and depends on the company, data quality, and modeling setup.

## IX. DISCUSSION

The results show that sentiment information can enhance stock trend prediction, but only under certain conditions. AAPL and TSLA benefit most clearly from sentiment integration. MSFT and AMZN show moderate model-dependent

improvements. GOOG and GOOGL show mixed effects, with sentiment helping only some models while reducing performance in others.

This pattern suggests that social media sentiment is not a universal predictor. Its usefulness depends on company-specific factors, investor behavior, public discussion intensity, and the degree to which online sentiment reflects market expectations. The results also show that influence weighting can be useful, especially for TSLA and selected MSFT models. This supports the hypothesis that who expresses sentiment matters, not only what sentiment is expressed.

The PageRank-weighted approach is valuable because it reduces the impact of low-visibility users and emphasizes accounts that occupy central positions in the Twitter interaction network. However, PageRank is not a perfect measure. It is computed on a fixed network and may not capture rapidly changing influence dynamics. Future systems should consider temporal influence measures.

## X. LIMITATIONS AND FUTURE WORK

This study has several limitations. First, the experiments focus on six large NASDAQ companies, so the findings may not generalize to smaller firms, other sectors, or other markets. Second, the Twitter data may contain noise, promotional content, spam, or bot-generated activity. Third, PageRank is calculated as a static network measure, while real-world influence changes over time. Fourth, stock prices are affected by macroeconomic events, earnings reports, interest rates, inflation, and sector-level shocks that are not fully captured in the current feature set.

Future work should extend the framework in several directions. First, the dataset should be expanded to include more companies and longer time periods. Second, sentiment from financial news sources such as Reuters, Bloomberg, CNBC, and company earnings reports should be integrated. Third, macroeconomic indicators such as inflation, interest rates, GDP growth, and unemployment should be included. Fourth, dynamic graph methods, temporal PageRank, Graph Neural Networks, and temporal Transformers could better model changing influence and market dynamics. Finally, financial Large Language Models could be used to extract richer event-level and reasoning-based signals from text.

## XI. CONCLUSION

This paper presented an influence-aware stock trend prediction framework combining Financial NLP, Social Network Analysis, and Machine Learning. A two-stage FinBERT adaptation strategy was used to classify financial tweet sentiment, first using VADER-generated weak labels and then manually annotated gold labels. The final model achieved 71.54% accuracy and 71.40% weighted F1-score.

The sentiment model was applied to approximately 3.6 million company-related tweets. A large Twitter interaction network was constructed, and PageRank was used to estimate user influence. Sentiment scores were then weighted

by PageRank and merged with historical stock prices and technical indicators.

The results show that sentiment features can improve stock trend prediction for selected companies. AAPL shows statistically significant improvement, TSLA benefits strongly from PageRank-weighted sentiment, and MSFT and AMZN show moderate model-dependent gains. However, GOOG and GOOGL show mixed results. The main conclusion is therefore nuanced: sentiment and user influence can improve prediction, but their effectiveness is company-dependent and should be evaluated carefully rather than assumed.

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